Machine Learning — End-to-End Workflow

Dataset → Data Check → Cleaning → EDA → X & y → Split → Preprocessing → Training → Prediction → Evaluation → Comparison → Best Model → Save / Deploy

1. Dataset Load

```
import pandas as pd
 df = pd.read_csv('data.csv')
```

2. Data Check

```
df.shape
 df.info()
 df.isnull().sum()
```

3. Data Cleaning

```
df = df.drop_duplicates()
 df = df.dropna()
```

4. EDA

```
df.describe()
 df['target'].value_counts()
```

5. X & y

```
X = df.drop('target', axis=1)
 y = df['target']
```

6. Train-Test Split

```
from sklearn.model_selection import train_test_split
 X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

7. Preprocessing

```
from sklearn.preprocessing import StandardScaler
 scaler = StandardScaler()
 X_train = scaler.fit_transform(X_train)
 X_test = scaler.transform(X_test)
```

8. Model Training

```
from sklearn.ensemble import RandomForestClassifier
 model = RandomForestClassifier(random_state=42)
 model.fit(X_train, y_train)
```

9. Prediction

```
y_pred = model.predict(X_test)
```

10. Metrics / Evaluation

```
from sklearn.metrics import accuracy_score
 print(accuracy_score(y_test, y_pred))
```

11. Model Comparison

```
Compare Accuracy / Precision / Recall / F1
 across multiple models.
```

12. Best Model

```
best_model = model  # highest validation performance
```

13. Save / Deploy

```
import joblib
 joblib.dump(best_model, 'best_model.pkl')
 # Load later: joblib.load('best_model.pkl')
```

Interview Flow: Load → Check → Clean → Explore → Split → Preprocess → Train → Predict → Evaluate → Compare → Select → Save → Deploy